

7 Electronic Design

7.1 General Design

In the Preliminary Design Review (PDR), we used a 4-board system with Linux and GPS modules separated. However, we concluded that interference between RF modules is lower than interference with the Linux board. We managed to integrate the GPS module onto the mainboard. The number of boards remains the same, but the GPS module is now on the mainboard.

1. **Mainboard** – System control, sensor reading, radio communication, logic
2. **Motor Control Board** – Actuator control
3. **Camera Board** – High-resolution image capture
4. **Linux Module** – Commercial Linux module

This architecture separates the control system (mainboard), high-speed camera/Linux operation, and high-voltage motor control. This reduces electrical interference between sub-systems and improves communication reliability.

7.1.1 Design Evolution and Component Selection

At the start of the project, multiple MCUs were evaluated: Atmel ATmega series (Arduino equivalent), STM32, ESP32, and RP2040.

- Atmel chips were rejected due to insufficient processing power and memory to handle multiple peripherals reliably.
- STM32 and ESP32 were considered but were more power-hungry, more complex, and exceeded the performance needs of Glidesat.
- RP2040 was selected due to its dual-core processing, multiple communication interfaces, low power consumption, and strong community support. Despite QFN package soldering difficulties, the performance-to-power ratio justified its selection.

7.1.2 Board Design

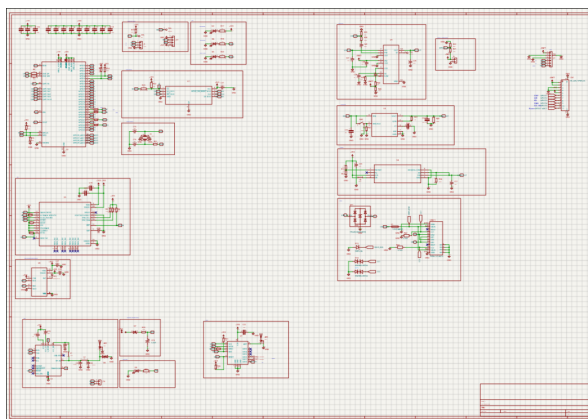


Figure 8: Schematic Design

The mainboard is a 4-layer PCB. Core processing is handled by the RP2040 SoC.

Complementary components:

- Flash memory – W25Q128JVS
- Crystal oscillator – X322512MSB4SI

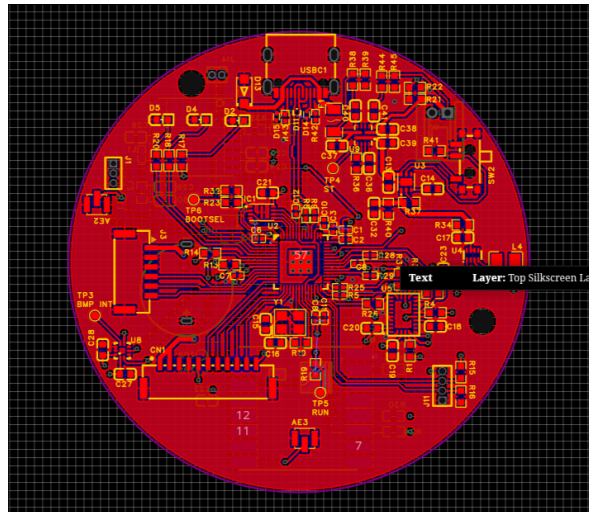


Figure 9: PCB Design

- Voltage regulator – AP2112K-3.3
- Power MUX – TPS2121RUXR
- Load switch – TPS22810DRVR
- Power MUX allows USB-C powering for flashing/testing.
- Load switch protects against high current draw which could damage a simple switch.

7.1.5 Camera/Linux Board

For imaging, the Luckfox Pico Mini and MIS5001 5MP Wide-Angle Camera Module were chosen.

- Raspberry Pi Compute Module 4 was replaced due to size and weight.
- Camera resolution was reduced to 5 MP due to the Pico Mini's 2-lane MIPI CSI limitation.
- High-resolution capture could not be done directly via RP2040 due to limited RAM, insufficient SPI bandwidth, and lack of MIPI CSI support.
- The Linux board triggers the camera via UART communication with the mainboard.

7.1.6 Motor Control Board

- Drives paraglider motors, regulates power for actuators, and includes a buzzer for recovery system (higher voltage requirement).
- Separation from mainboard prevents high-current electrical noise from affecting sensors or LoRa modules.
- Micro metal gear motor with encoder controlled via DRV8833 motor driver.

7.2 Power Supply

7.2.1 Power Source

Glidesat is powered by a rechargeable LiPo battery. Two options were considered: single-cell (3.7 V nominal) and dual-cell (7.4 V nominal).

- Dual-cell reduces current for the same power, improving wiring efficiency and allowing efficient buck conversion.
- Design considerations for dual-cell include charging circuit complexity and over-voltage protection.
- A dual-cell 2700–3000 mAh battery was selected to cover a 4-hour mission with safety margin (20 Wh total energy requirement).

7.2.2 Power Regulation

- Most electronics operate at 3.3 V (RP2040 and sensors) using a high-efficiency buck converter.
- Battery voltage monitored via mainboard ADC.
- Distribution via JST GH connectors to prevent disconnection.
- 3.3 V for mainboard components, raw 7.4 V to motor and Linux boards; 5 V regulated onboard for camera and motors.

7.3 Communication System

Glidesat communicates with the ground station via RFM95W-868S2 LoRa transceiver. The system supports bi-directional communication for telemetry and command reception.

Data transmitted from ;'[]:

- Altitude
- Temperature
- GPS coordinates
- Battery voltage

- System status
- Camera status

Data received by Glidesat:

- Motor direction
- Motor speed
- Recovery system activation